

1. Existing Use Case

Thermo Fisher's Product Family Pages (PFP) and Product Detail Pages (PDP) in the commerce/PDP system embed **content pods** — targeted HTML fragments authored and managed in AEM, delivered into non-AEM commerce pages at runtime.

Sample URLs (Live)

2. Current Flow

```

1 https://www.thermofisher.com/in/en/product-family-syndicated-
  content.pfpsnippet.html/sku/CHROMELEON7/3
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```

URL Part	Value	Purpose
/in/en/	Country=India, Language=English	Locale for content resolution
product-family-syndicated-content	AEM page node	Entry point — acts as servlet trigger only

<code>.pfpsnippet</code>	Sling selector	Triggers the syndication servlet
<code>.html</code>	Extension	Standard AEM HTML extension
<code>/sku/CHROMELEON7/3</code>	Suffix	type=sku, value=CHROMELEON7, pod=3

2.2 Component Architecture in AEM

Content Snippet Template (`lifetech/templates/content_snippet`)

- Specialized AEM page template for fragment pages
- Page component: `lifetech/components/pages/content-snippet`
- Renders via `content-snippet.jsp`
- On publish: outputs raw HTML with no page shell
- Uses a custom paragraph system: `SnippetParsys`

SnippetParsys (`lifetech/components/snippetparsys`)

- Custom AEM paragraph system (container)
- Supports column layouts (START/BREAK/END)
- Strips decoration tags on publish for clean HTML output
- Allowed components: text, textimage, heading, Brightcove video, raw HTML, interactive containers

Tag Taxonomy

- Namespace: `pfp:` (Product Family Page)
- Tag format: `pfp:sku/{SKU-VALUE}/pod_{POD-ID}`
- Example: `pfp:sku/CHROMELEON7/pod_3`
- Authors tag each Content Snippet page with the appropriate tag to register it for a SKU + pod combination

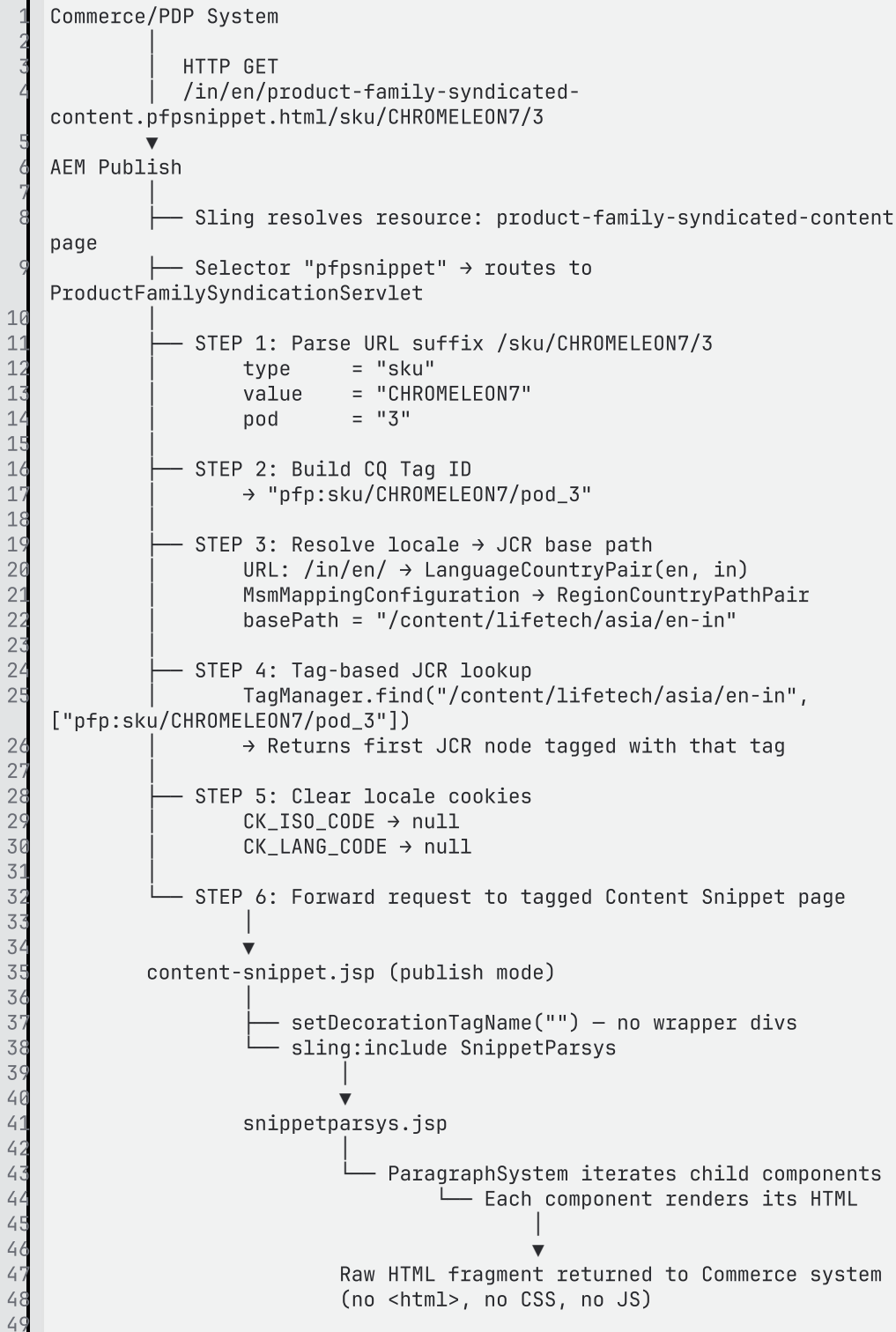
2.3 Servlet — ProductFamilySyndicationServlet

File: `lifetech-web-core/src/main/java/com/lifetechnologies/web/servlets/ProductFamilySyndicationServlet.java`

Registration:

```
1 sling.servlet.resourceTypes = sling/servlet/default (any resource)
2 sling.servlet.selectors     = pfpsnippet
3 sling.servlet.extensions    = html
4 sling.servlet.methods       = GET
5
```

2.4 End-to-End Request Flow



3. Proposed EDS Architecture

3.1 Core Principles

The EDS migration replaces the entire AEM-side syndication stack with three components:

1. **DA (da.live)** — authors create and manage fragment content (replaces AEM Sites + Content Snippet Template)
2. **EDS blocks** — define the rendering and styling of content components (replaces JSP components + SnippetParsys)
3. `aem-embed.js` — a Web Component that fetches and renders fragments in the commerce page (replaces the Java servlet + `content-snippet.jsp` pipeline)

3.2 The `aem-embed` Web Component

Adobe provides an open-source Web Component specifically designed for this pattern:

Repository: `https://github.com/adobe/aem-embed`

Documentation: `https://www.aem.live/docs/aem-embed`

`aem-embed.js` is a standards-based **Custom Element** (`<aem-embed>`) that:

1. Reads the `url` attribute
2. Automatically appends `.plain.html` to fetch the EDS fragment
3. Creates an isolated **Shadow DOM** — prevents CSS/JS conflicts with the commerce page
4. Loads `styles/styles.css` from the EDS repo into the Shadow DOM
5. Runs EDS's `decorateMain()` pipeline to convert DA table markup into block structure
6. For every block in the fragment, loads its dedicated `blocks/{name}/{name}.css` and `blocks/{name}/{name}.js` into the Shadow DOM
7. Renders fully styled, fully functional content — isolated from the commerce page

3.3 How Commerce System Uses It

One-time setup — add script to commerce /PDP page `<head>` :

```
1 <script src="https://www.thermofisher.com/scripts/aem-embed.js"
2   type="module"></script>
```

Per-product — drop tag where each pod should appear:

```
1 <aem-embed
   url="https://www.thermofisher.com/in/en/snippets/sku/CHROMELEON7/pod-
   1"></aem-embed>
2 <aem-embed
   url="https://www.thermofisher.com/in/en/snippets/sku/CHROMELEON7/pod-
   2"></aem-embed>
```

```

3 <aem-embed
  url="https://www.thermofisher.com/in/en/snippets/sku/CHROMELEON7/pod-
  3"></aem-embed>
4

```

The commerce system constructs the URL from data it already owns: locale + SKU + pod number.

3.4 EDS Repo Structure

```

1 TFS EDS Repository
2 |— scripts/
3 |   |— aem-embed.js      ← copied from adobe/aem-embed (Adobe-
  maintained)
4 |   |— scripts.js        ← standard EDS scripts, exports
  decorateMain()
5 |   |— styles/
6 |       |— styles.css    ← global EDS styles (loaded into Shadow
  DOM)
7 |       |— fonts.css     ← fonts (loaded into Shadow DOM)
8 |   |— blocks/
9 |       |— text/
10 |           |— text.js
11 |           |— text.css
12 |       |— textimage/
13 |           |— textimage.js
14 |           |— textimage.css
15 |       |— video/
16 |           |— video.js
17 |           |— video.css
18 |       |— ...
19 |   (DA content synced by EDS pipeline)
20 |   /in/en/snippets/sku/CHROMELEON7/pod-1
21 |   /in/en/snippets/sku/CHROMELEON7/pod-2
22 |   /us/en/snippets/sku/CHROMELEON7/pod-1
23

```

3.5 DA Authoring

Authors work in **da.live** . Each fragment is a document saved at the URL path convention:

```

1 /in/en/snippets/sku/{SKU}/pod-{N}
2

```

Inside the document, authors use standard EDS block syntax:

```

1 +-----+
2 | text                                     |
3 +-----+
4 | ## Chromeleon Software                 |
5 | Increase productivity...               |
6 | [Learn more](url)                     |
7 +-----+
8
9 +-----+
10 | video                                  |
11 +-----+
12 | https://brightcove.../xyz             |
13 +-----+
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```

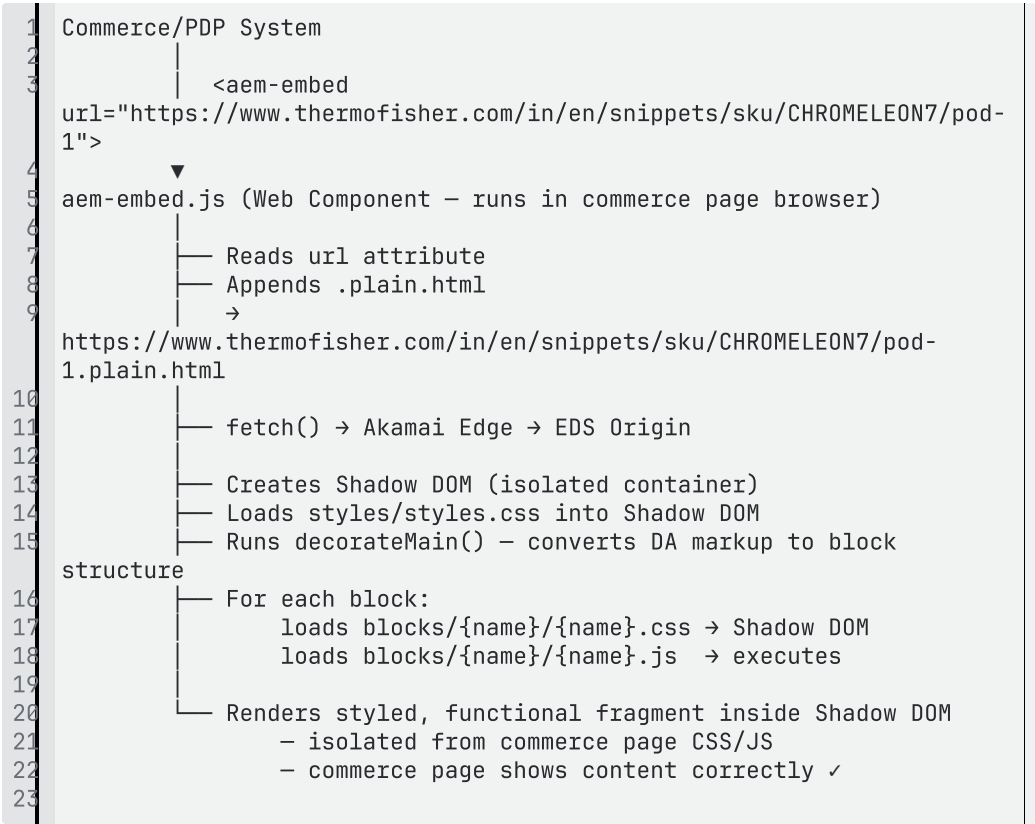
No tags. No taxonomy. No template selection. Just a document at the right path.

3.6 CORS Configuration

Because the commerce system and the EDS content are served from the same origin (`thermofisher.com`), no CORS configuration is required. If the commerce system is hosted on a different domain/origin, then configure CORS in the TFS EDS project headers config:

```
1 | access-control-allow-origin: *
2 |
```

3.7 End-to-End Request Flow (EDS)



4. Impact Analysis

4.1 Commerce / PDP System — Changes Required

Change	Details
Add <code>aem-embed.js</code> script tag	One <code><script></code> tag in page <code><head></code>
Replace servlet HTTP calls with <code><aem-embed></code> tags	Replace existing GET calls to <code>.pfpsnippet.html/sku/X/N</code>

	with <code><aem-embed url="..."></code> tags
URL construction logic	Build EDS URL from existing data: <code>/ {locale} / snippets / sku / {SKU} / pod - {N}</code>
CSS updates	Commerce page CSS may need updates to work with Shadow DOM boundaries. Global styles that targeted injected AEM HTML structure will need review

4.2 TFS AEM — What Is Decommissioned

AEM Artifact	Status
<code>ProductFamilySyndicationServlet.java</code>	Decommissioned
<code>MsmMappingConfiguration.java</code>	Decommissioned
<code>content-snippet.jsp</code>	Decommissioned
<code>snippetparsys.jsp</code> + <code>snippetparsys</code> component	Decommissioned
<code>content_snippet</code> AEM template	Decommissioned
<code>pfp:</code> tag namespace and taxonomy	Decommissioned
AEM MSM live copies for snippet pages	Decommissioned (replaced by DA MSM)
All Content Snippet pages in AEM Sites	Migrated to DA

5. Runtime Ownership

5.1 Adobe Development Team

Responsibility	Task
EDS repo setup	Initialise TFS EDS project with standard <code>scripts/scripts.js</code> , <code>styles/styles.css</code>
Copy <code>aem-embed.js</code>	Copy from <code>github.com/adobe/aem-embed</code> into <code>/scripts/aem-embed.js</code>
Build EDS blocks	Develop <code>text</code> , <code>textimage</code> , <code>video</code> , <code>heading</code> blocks with JS + CSS
CORS configuration	Add <code>access-control-allow-origin: *</code> to EDS headers config if domain differs
DA MSM setup	Configure base site + satellite site relationships in DA org config
URL convention definition	Define and document the <code>/snippets/sku/{SKU}/pod-{N}</code> path convention

5.2 TFS Content Team (Authors)

Responsibility	Task
Path convention adherence	Save documents at <code>/in/en/snippets/sku/{SKU}/pod-{N}</code> — path IS the identifier
No tagging required	Tags are fully eliminated — authors do not need to manage any tag taxonomy

5.3 Commerce / PDP Team

Responsibility	Task
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Script integration	Add <code>aem-embed.js</code> script tag to commerce page template
Tag integration	Replace existing servlet HTTP calls with <code><aem-embed url="..."></code> tags
URL construction	Build EDS URL from locale + SKU + pod data available in commerce system
CSS review	Review and update any commerce-side CSS that targets AEM-specific HTML structure

6. Parity Check

6.1 Functional Parity

Capability	AEM (Current)	EDS + DA (Target)	Parity
Rich content authoring (text, images, video)	AEM Sites — drag & drop components	DA — block authoring	✅ Equivalent
Raw HTML fragment delivery	<code>content-snippet.jsp</code> strips page shell	<code>.plain.html</code> returns <code><main></code> content only	✅ Equivalent
Styled content delivery	Commerce page CSS styles AEM HTML	<code>aem-embed.js</code> loads block CSS into Shadow DOM	✅ Improved — TFS now owns styling
Locale-specific content	AEM MSM live copies per locale	DA MSM satellite sites per locale	✅ Equivalent
Multiple pods per SKU	Multiple tagged pages per SKU	Multiple DA documents per path convention	✅ Equivalent

SKU-based content lookup	Tag taxonomy <code>pfp:sku/{SKU}/pod_{N}</code>	URL path <code>/snippets/sku/{SKU}/pod-{N}</code>	✓ Simplified
Commerce system independence	Commerce calls AEM servlet URL	Commerce uses <code><aem-embed></code> tag	✓ Improved — cleaner contract
Content updates without commerce deploy	Yes — AEM publish	Yes — DA publish	✓ Equivalent
Block JS initialisation (video players etc.)	AEM clientlibs per component	<code>aem-embed.js</code> loads block JS per block	✓ Equivalent
CSS isolation from commerce page	No isolation — commerce CSS must not conflict	Full Shadow DOM isolation	✓ Improved

6.2 Technical Parity

Technical Concern	AEM (Current)	EDS + DA (Target)
Servlet / entry point	<code>ProductFamilySyndicationServlet.java</code> (Java OSGi)	<code>aem-embed.js</code> (Web Component, browser-side)
Content lookup mechanism	<code>TagManager.find()</code> — JCR tag search	URL path convention — no lookup needed
Locale resolution	<code>MsmMappingConfiguration.java</code> — server-side Java	DA MSM — platform-level, at delivery time
Rendering pipeline	<code>content-snippet.jsp</code> +	EDS <code>decorateMain()</code> + block decoration

	snippetparsys.js p	
Maintenance	Java OSGi bundle — requires AEM developer	Standard JS/CSS — any front-end developer

7. Tag Removal and URL Simplification

7.1 URL Convention Replaces Tags Entirely

In EDS + DA, the **path where the author saves the document IS the identifier**. No separate tagging step. No taxonomy to manage.

The rule is simple:

```
1 /{locale}/snippets/sku/{SKU-VALUE}/pod-{POD-NUMBER}
2
```

Examples:

Content	DA Document Path
CHROMELEON7, Pod 1, India English	/in/en/snippets/sku/CHROMELEON7/pod-1
CHROMELEON7, Pod 2, India English	/in/en/snippets/sku/CHROMELEON7/pod-2
CHROMELEON7, Pod 1, US English (base)	/us/en/snippets/sku/CHROMELEON7/pod-1
CHROMELEON7, Pod 1, German	/de/de/snippets/sku/CHROMELEON7/pod-1
NEWPRODUCT123, Pod 1, US English	/us/en/snippets/sku/NEWPRODUCT123/pod-1

7.2 URL Format Comparison

	AEM (Current)	EDS + DA (Target)
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Commerce calls	<code>/ {locale} /product -family- syndicated- content.pfpsnippe t.html/sku/{SKU}/ {POD}</code>	<code><aem-embed url="/ {locale} /sn ippets/sku/{SKU}/ pod-{N}"></code>
Example	<code>/in/en/product- family- syndicated- content.pfpsnippe t.html/sku/CHROME LEON7/3</code>	<code>url="/in/en/snipp ets/sku/CHROMELEO N7/pod-3"</code>
Content identifier	Tag: <code>pfp:sku/CHROMELEO N7/pod_3</code>	Path: <code>/in/en/snippets/s ku/CHROMELEON7/po d-3</code>
Lookup mechanism	JCR TagManager search	Direct URL resolution
Human readable	No	Yes